

Technology Validation and Start Up Fund

Round 36 Proposal Evaluations

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TECHNOLOGY VALIDATION AND STARTUP FUND

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1) Executive Summary

Redwood is a Columbus, Ohio based LLC founded by former Battelle executives over 11 years ago. Redwood has assembled an extraordinary team for this Program. Each member of the five-person Redwood team is an accomplished technology commercialization professional with decades of experience in performing business and technical evaluations. This team, combined with identified external subject matter experts, has extensive experience in all six of the Ohio Third Frontier technology focus areas. More detail on the Redwood team is provided in Appendix 1 of this report and on our website (www.Redwdinnnov.com). Details of the TVSF program and the review process are provided in Appendix 2.

Eighteen (18) TVSF Round 36 applications, totaling \$3,450,000, were received and evaluated.

The proposals consisted of 3 Phase 1 application totaling \$500,000 (1 Phase 1, Option 3 applicant of \$100,000 and 2 Phase 1, Option 1 applicants of \$200,000 each) and 15 Phase 2 applications totaling \$2,950,000 for a combined total of \$3,450,000.

Funding is recommended for two Phase 1 applications totaling \$300,000 and twelve Phase 2 applications for a total of \$2,350,000 yielding a combined total of \$2,650,000. This translates to a 78% recommended application funding rate for Phase 1/Phase 2 TVSF round, compared to the average of 48% over all 36 TVSF rounds. The recommended approval rate for Phase 2 proposals in this round is 80%.

2) Evaluation Results

Summaries of the evaluations of the proposals for Round 36, incorporating the interview, and funding recommendations are shown below in Tables 1 and 2. Rounds 22-29 interviews were conducted via video due to COVID-19 concerns. Round 30 interviews were conducted in person and subsequent Rounds were conducted via video. The total recommended funding for Phase 1 and Phase 2 projects is, respectively, \$300,000, and \$2,350,000. Note that the Table 1 and 2 column widths are proportional to the weighting of the evaluation criteria. For example, in Table 1, Selection Committee which is weighted at 20 is four times as wide as project selection which is weighted at 5. Note that a yellow evaluation indicates that the proposal meets that particular criterion.

More detailed evaluations and recommendations for each Phase 1 and Phase 2 proposal may be found in Section 3 of this report.

Table 1 – Phase 1 Proposal Evaluation and Funding Recommendation

Table 1
Phase 1 Proposal Evaluation and Funding Recommendation
TVSF Round 36

Proposal #	Applicant	Requested Funding (\$1,000)	Recommended Funding (\$1,000)	Selection Committee	Deal Flow & Budget Strategy	External Participation	Track Record	Metrics	Project Management & Experience	Project Selection Process
FY24-8312	Research Institute of Nationwide Childrens Hospital	200	200							
FY24-8313	University of Akron Research Foundation Spark Fund	100	100							
FY24-8314	Cleveland State University	200	-							
	Total	500	300	Column width is proportional to score weighting in each category						
Evaluation Scale				Weak		Meets		Exceeds		

- Option 1:** \$200,000 to \$500,000, 1:1 cost share, 1 year to allocate funds, 2 year project period.
- Option 2:** \$1,000,000, 1:1 cost share, 2 years to allocate funds, 3 year project period.
- Option 3:** \$100,000, internship component, 1 year to allocate funds, 2 year project period.

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Table 2 – Phase 2 Proposal Evaluation and Funding Recommendation

	Proposal Number	Lead Applicant	Requested Funding (\$1,000)	Recommended Funding (\$1,000)	Team	Opportunity/Market Size	IP Protection	Proof of Concept	Potential Investor & Business Partner Engagement	Business Model	Project Plan/Budget Narrative	Growth Plan in Ohio	ESP Interaction
	FY24-8316	Reglia Therapeutics	200	200									
	FY24-8317	CapV	200	200									
	FY24-8318	Nabu Optical Systems	200	200									
	FY24-8321	Auxilium Health	200	200									
	FY24-8322	Kalix	200	200									
	FY24-8323	NeoIndicate	150	150									
	FY24-8324	Spectin	200	200									
	FY24-8325	Afference	200	200									
	FY24-8326	Visual Detection Systems	200	200									
	FY24-8327	Argos Advanced Solutions	200	200									
	FY24-8328	Lyceum AI	200	200									
	FY24-8329	Engineered Products of Ohio	200	200									
	Sub-Total		2,350	2,350									
	FY24-8315	Protein Capture Science	200	-									
	FY24-8319	CrossLIMS	200	-									
	FY24-8320	Shark	200	-									
	Sub-Total		600	-	Column width is proportional to score weighting in each category								
	Total		2,950	2,350									
	Evaluation Scale				Weak		Meets		Exceeds				

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Table 3 lists the funding approval rate by TVSF round. This round's approval rate of 78% of the total submitted proposals is on the high side of the average over all rounds. The Phase 1 applications recommended for approval comprised 14% of the submittals for this round. The historical range of recommendations for individual rounds has spanned 27 – 100%, with an average of 48%.

Table 3. TVSF Approval Rate by Round

TVSF Round 36 Approval Rate by Round Phase 1/Phase 2

Round	\$ Recommended	Approval Rate
1 (APR 2012)	\$950,000	35%
2 (AUG 2012)	\$900,000	52%
3 (DEC 2012)	\$610,000	44%
4 (JUN 2013)	\$864,000	30%
5 (FEB 2014)	\$1,462,000	46%
6 (JUN 2014)	\$998,000	39%
7 (OCT 2014)	\$1,100,000	57%
8 (FEB 2015)	\$710,000	37%
9 (JUN 2015)	\$550,000	31%
10 (DEC 2015)	\$925,000	38%
11 (APR 2016)	\$1,239,000	46%
12 (OCT 2016)	\$3,537,269	46%
13 (MAR2017)	\$1,567,500	38%
14 (SEP 2017)	\$498,832	27%
15 (DEC 2017)	\$2,250,000	38%
16 (MAR 2018)	\$2,098,600	52%
17 (SEP 2018)	\$2,100,000	42%
18 (DEC 2018)	\$1,150,000	35%
19 (APR 2019)	\$2,250,000	43%

Round	\$ Recommended	Approval Rate
20 (NOV 2019)	\$1,350,000	43%
21 (FEB 2020)	\$3,944,000	56%
22 (JUN 2020)	\$1,398,630	53%
23 (DEC 2020)	\$900,000	50%
24 (MAR 2021)	\$2,092,900	55%
25 (JUN 2021)	\$800,000	75%
26 (OCT 2021)	\$1,700,000	55%
27 (FEB 2022)	\$850,000	43%
28 (APR 2022)	\$2,499,976	64%
29 (JULY 2022)	\$850,000	100%
30 (OCT 2022)	\$3,700,000	71%
31 (JAN 2023)	\$100,000	50%
32 (APR 2023)	\$850,000	64%
33 (JULY 2023)	\$1,100,000	73%
34 (OCT 2023)	\$250,000	33%
35 (JAN 2024)	\$800,000	59%
36 (APR 2024)	\$2,650,000	78%
Total Funding	\$51,595,707	
Average/Round	\$1,433,214	48%

3) Proposal Summaries

Phase 1 Proposal Summaries - Recommended for Funding

Proposal 24-8312	The Research Institute at Nationwide Children's Hospital	Amount Requested: \$200,000
Prior Phase 1 Applications: Yes	Ohio Third Frontier Validation & Start-Up Fund (Round 36)	Amount Recommended: \$200,000

TOTAL BUDGET: \$400,000

Institution Snapshot: The strategic vision of the program is to advance inventions/intellectual property along the commercialization pathway with the purpose of making early inventions both more attractive and less risky to potential licensing and/or business partners.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Selection Committee	Competent 4-member committee, all with evaluation experience and several with commercialization of early-stage life science expertise.
G	Deal Flow; Budget Strategy	There were 80 invention disclosures in 2023 and 22 licensing deals in the same year. TDF pipeline focuses on 2 areas- biotechnologies and end user innovations.
G	External Participation	External providers will be utilized when appropriate. There is a recognition that some activities may need to be performed in house if the project would be jeopardized by using outside vendors.
G	Track Record	Since 2010, 62 (52 TDF/10 TVSF) of 149 TDF/TVSF applications have been funded resulting in 49% licensed or pending licensing deals.
G	Metrics	10 TVSF projects funded, 5 licensed to startups or optioned or in license negotiations, 2 being marketed with licensing interest, 1 project in prep, 1 project ongoing, 1 project was closed.
G	Project Management & Experience	The Office of Technology Commercialization manages the program and the licensing associate managing the inventor review interim and final reports.
G	Project Selection	Clear process with TVSF committee members in prior OTC review.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: The program at Nationwide Children's Hospital continues to produce exceptional outcomes for the Institute. Numerous licensing and commercialization opportunities of the intellectual property have provided new treatments and better outcomes for pediatric patients.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 24-8313	University of Akron Research Foundation	Amount Requested: \$100,000
Prior Phase 1 Applications: Yes	<i>University of Akron Research Foundation Spark Fund</i>	Amount Recommended: \$100,000

TOTAL BUDGET: \$100,000

Institution Snapshot: The University of Akron Research Foundation (UARF) Spark Fund will provide the proof needed to bring University of Akron (UA) technologies in areas of institutional strength to the point where a scalable startup can be built based on the technology.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Selection Committee	Six-member selection committee backgrounds and market expertise are well aligned to UARF strengths.
G	Deal Flow; Budget Strategy	UARF anticipates 10 applicant Project pitches, 4-5 projects in due diligence and 2-3 projects funded.
Y	External Participation	A creditable and generic external provider plan is provided. Funds are held by UARF and disbursed to third party providers to complete tasks.
G	Track Record	Three Spark Funds have resulted in 12 funded projects with 5 license 1 exit, 2 startups.
G	Metrics	Outcome metrics are well considered and appropriate.
G	Project Management & Experience	Established 4 step process to identify and select projects for funding is well considered and appropriately leverages Senior Fellows. Expanding Senior Fellows team.
G	Project Selection	
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: UARF continues to identify appropriate level intellectual property and use TVSF Phase 1 funds to create value that can be leveraged with startups and other licensees. The concept of Senior Fellow to help with the identification of the need in the market place coupled with guidance offered as the project is executed provides significant value to the project team.

Phase 1 Proposal Summaries - Not Recommended for Funding

Proposal 24-8314	Cleveland State University	Amount Requested: \$200,000
Prior Phase 1 Applications: Yes	<i>TVSF TeCK Fund</i>	Amount Recommended: \$0
TOTAL BUDGET: \$0		

Institution Snapshot: Cleveland State University and Kent State University are proposing to continue a unique joint Phase 1 TVSF program to accelerate commercial development of technologies originating within each institution.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Selection Committee	The background, experience and diverse perspectives of the committee is appropriate.
Y	Deal Flow; Budget Strategy	Deal flow is limited but assumed to be high quality. Further promoting the program to encourage added deal flow is recommended.
Y	External Participation	A credible list of outside service providers is given, spanning several product / market spaces.
R	Track Record	The track record over all three TeCK funds is a significant concern. The rate of licensing and attracted investment is quite small relative to other Phase 1 programs.
Y	Metrics	The outcome-based metrics are appropriate. It is encouraged that leading indicator type metrics also be instituted, if not already in use.
Y	Project Management & experience	The recent adoption of a more active project management approach described in the proposal is good to see.
Y	Project Selection	Given the track record, project selection criteria should be reviewed.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: Given the track record of the 3 TeCK funds, this application is not recommended for funding. The CSU/KSU team is encouraged to reapply in six months when the positive results of their improvement plan will be apparent.

Phase 2 Proposal Summaries - Recommended for Funding

Proposal 24-8316	Reglia Therapeutics, Inc.	Amount Requested: \$200,000
<i>Licensing Institution</i>	Case Western Reserve University	Amount Recommended: \$ 200,000
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: No	<i>Astrocyte medicines for neurodegenerative diseases</i>
Biomedical/ Life Sciences		JumpStart

Company Snapshot: Reglia Therapeutics is evaluating a new method to treat neurodegenerative disease and patient disability by controlling toxic astrocytes, the support cells of the brain.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Opportunity/Market Size	Large growing market covering many neurological diseases. Alzheimer's market in the U.S. alone is \$22.6 billion.
G	Management Team	Experienced and well-established scientific co-founders. Founding scientist, EIR and business lead, all have a successful track record and co-working history. Leadership team is experienced at executing their business model.
Y	Intellectual Property Protection	Two pending applications covering their proprietary discovery platform. Each drug candidate will be patented as composition of matter.
G	Proof of Concept	Successful TVSF Phase 1 project. Technology validated in a proof-of-concept drug screen resulting in suppression of toxic astrocyte formation and promoting neuroprotection in <i>in vivo</i> mouse models.
G	Potential Investor/ Business Partner Engagement	In discussions with angel investors, Cleveland Life Science Accelerator Fund, and Ohio Innovation Fund for \$1M in 2024.
G	Business Model	Model is to use their proprietary screening platform as an "engine" to discover and develop new drugs for neurologic diseases.
Y	Project Plan/ Budget Narrative	Clear objectives, achievable with providers listed, time & budget.
Y	Growth Plan in Ohio	CWRU technology, incubated in Ohio, with Ohio subs and investors
G	ESP Interaction	Engaged with JumpStart and Bounce Innovation Hub
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Highly experienced team that has successfully launched companies together over the last decade and a half, one of which has raised \$85M. Very large market opportunity and need for effective treatments. The technology platform was successfully demonstrated in a TVSF Phase 1 project. This application is recommended for funding.

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Proposal 24-8317	CapV LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	University of Dayton	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	CapV - A 5G Microelectronics Company
Advanced Manufacturing		Entrepreneurs' Center

Company Snapshot: CapV's varactor (variable capacitor) is a tunable thin film element that enables digital tuning of radio frequency filters to avoid interfering signals and data loss to provide a secure communication link at a higher frequency and bandwidth.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	A small but very effective Team with proven record in the field and good connections with DARPA and Lockheed Martin supplier. Active inputs from ESP and Local business incubator staff.
G	Opportunity/Market Size	TAM is over \$ 100B/yr. SAM is \$ 10B/yr. SOM is \$ 400 MM/yr. with 30 % CAGR.
Y	Intellectual Property Protection	4 older patents from 2006-2015 mentioned with some undisclosed trade secrets.
Y	Proof of Concept	The basic concepts/ prototypes were developed in a DARPA Phase I/II \$ 6M funding under WRAP program that ended recently. Currently is a TRL 5.
Y	Potential Investor/ Business Partner Engagement	Actively exploring investments from Lockheed Martin (LM) ventures.
Y	Business Model	Seems well laid; Capturing 5 % of SOM, revenue projection is \$ 16 MM in 2028.
Y	Project Plan/ Budget Narrative	A well laid out project plan and budget narrative.
Y	Growth Plan in Ohio	15-20 staff projected in 2028
Y	ESP Interaction	Active interactions with the Entrepreneurs' Center.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: A very strong technical Team with proven track record in the field with DARPA, Lockheed Martin and other suppliers. Proactive engagements with ESP and local business incubator to reach out to potential investors once the TVSF project has shown > 70 % yield. The Team is strongly urged to add other staff/ capabilities needed for scale-up and manufacturing and in house business development skill set.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 24-8318	Nabu Optical Systems, LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	University of Dayton	Amount Recommended: \$ 200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	Anu: An Advanced Manufacturing Unit for Nanofabrication
Advanced Manufacturing		Entrepreneurs' Center

Company Snapshot: Nabu is developing an affordable optical nanofabrication prototype for a 'lab to fab' path for creators of new chip designs which provides higher throughput and resolution.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	A well-balanced Team with Technical and Business development skill sets including fund raising track record.
Y	Opportunity/Market Size	Large global market in the \$ billions/ yr. U.S Target market size in the > \$100M/ yr.
Y	Intellectual Property Protection	Patent application 63/550745 has been filed. Working with the University of Dayton to secure an exclusive license.
G	Proof of Concept	Alpha prototype developed and reached a TRL of 3/4. Results show that the prototype exceeds current competitive products in key performance metrics. Beta version developed.
G	Potential Investor/ Business Partner Engagement	Discussions with Rushlight Ventures and others equity investment in 2025 and 2026.
G	Business Model	A well-articulated business model with projected revenues >\$ 22M in 2028. Both direct and contract manufacturing sales are envisioned.
Y	Project Plan/ Budget Narrative	A well laid out project plan and budget narrative. TRL 5/6 after TVSF.
Y	Growth Plan in Ohio	Growing from 5-6 to over 50 in 2028
Y	ESP Interaction	Close interaction with Entrepreneurs' Center.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: A very strong Team with excellent technical track record and connections with target customers for the initial product launch and subsequent entry to larger markets. Good connections with investment community. Working closely with ESP to access needed capabilities at this early stage. The Team is strongly urged to start adding staff/ capabilities soon to fully realize the projected business revenues.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 24-8321	Auxilium Health, Inc.	Amount Requested: \$200,000
<i>Licensing Institution</i>	University of Akron Research Foundation	Amount Recommended: \$200,000
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: No	<i>Dual Layer Smart Wound Dressing for Chronic Wounds</i>
Biomedical/ Life Sciences		Bounce Innovation Hub; JumpStart

Company Snapshot: Auxilium's aerogel technology provide layered dressings that offer a unique method to aid in the healing of wounds.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	The company is led by two impressive recent PhDs and advised by multiple physicians and health care veterans. An experienced medical device salesperson will be added to the team as funding becomes available.
Y	Opportunity/Market Size	There is a large market for improved wound treatment approaches. Encouragingly, a precedent has been established of premium pricing for advanced products.
Y	Intellectual Property Protection	The company has filed a PCT application in December 2023 that covers the key product attributes.
Y	Proof of Concept	The product concept addresses an important unmet need. The product is modular and smart. In-vivo demonstration of the MVP is the goal of the TVSF project.
Y	Potential Investor/ Business Partner Engagement	The team has had significant engagement with clinicians and early-stage investors.
Y	Business Model	A modular product design leads to recurring revenue and early market introduction. Reimbursement is well understood.
Y	Project Plan/ Budget Narrative	The project plan is appropriate, the budget is optimistic but possible.
G	Growth Plan in Ohio	The company is committed to growing in Ohio.
Y	ESP Interaction	Interaction with JumpStart and others in NE Ohio ecosystem is clear.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: This application is recommended for funding. Auxilium has developed a novel product that addresses an important unmet need in wound care. The young team is impressive, coachable and understands the market.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 24-8322	Kalix	Amount Requested: \$200,000
<i>Licensing Institution</i>	University of Dayton	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>Intelligent System for Farm Disease Management</i>
Biomedical/ Life Sciences		Entrepreneurs' Center

Company Snapshot: Kalix is developing a low-cost, real-time, and on-site disease management system tailored for agriculture, food and meat producers that provides testing results in one hour.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	The three-person team has technical, startup, manufacturing, market and fundraising experience on the team.
Y	Opportunity/Market Size	Kalix is initially focusing on the cattle market in the US. The technology is a platform that can be adapted into various livestock and crop assessment testing protocols.
Y	Intellectual Property Protection	A utility patent application was filed on March 16, 2023 with priority date of March 16, 2022. Additional patents are anticipated to be filed in the future.
Y	Proof of Concept	The concept has been tested with both Listeria and Covid-19 in 2022. Additional customization is required to prove testing for the cattle disease market and other markets to be developed in the future.
Y	Potential Investor/ Business Partner Engagement	Company has raised \$25K for 5% equity. If accepted into the AgLaunch program, will receive another \$125K for 5% equity.
Y	Business Model	A razor/razor blade business model with significant potential for growth.
Y	Project Plan/ Budget Narrative	Budget is consistent to build base units & test kits for AgLaunch trial.
Y	Growth Plan in Ohio	Kalix estimates that there will be 40 manufacturing jobs in 2028.
Y	ESP Interaction	Working with the Entrepreneurs' Center in Dayton.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Kalix has developed a test kit with base units to analyze the samples for the detection of various diseases. The unique feature of this technology is the ability to use it in the field and obtain results within 1 hour. Current practice is to send samples out to a lab for testing. Results are generally available in 2 to 5 business days. If this platform is successful it will provide for rapid results for not only cattle, but other animals to detect disease quickly and be able to treat or dispose of animal.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 24-8323	NeolIndicate, LLC	Amount Requested: \$150,000
<i>Licensing Institution</i>	Case Western Reserve University	Amount Recommended: \$150,000
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: 25,33	Radiotheranostics for Brain Tumors
Biomedical/ Life Sciences		JumpStart; Rev1

Company Snapshot: NeolIndicate is developing theranostic (therapeutic and diagnostic) products that can provide rapid, accurate information for clinicians to utilize a “detect and treat” strategy for diagnosing, imaging and selectively treating a variety of cancer types.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	Strong leadership team, credible in areas of expertise needed to achieve technical and business milestones and progress toward profitability.
G	Opportunity/Market Size	Potential market (US) is \$4.5B for a single cancer. Subsequent products potentially useful in prostate, lung, melanoma, ovarian and breast cancer increases market opportunity to >\$200B.
G	Intellectual Property Protection	Seven US issued patents, one recently allowed and others pending. License for new IP and new field of use is currently being negotiated.
Y	Proof of Concept	Compound has been demonstrated in mouse models and human tissue samples.
G	Potential Investor/ Business Partner Engagement	Established relationships with angel and venture investors. The company has raised equity funding of \$560,000 from those sources.
Y	Business Model	The company will generate high value products based on its platform technology. Focus is on early pre-clinical and clinical development.
Y	Project Plan/ Budget Narrative	Use existing cGMP materials to produce materials and test in mice.
Y	Growth Plan in Ohio	Plans provide for growth in Ohio and leverage existing ecosystem
G	ESP Interaction	Engaged with multiple ESPs: JumpStart, CincyTech, Rev1
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: NeolIndicate proposes to develop a PET scan and radiotherapy agent based on new IP and new field of use. The products generated, based upon the prototype compound, could improve cancer treatment and earn significant revenue. Inventor/ founder is the central team member whose continued participation is essential. This application is recommended for funding.

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Proposal 24-8324	Spectrin Inc.	Amount Requested: \$200,000
<i>Licensing Institution</i>	Ohio State University	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 22,35	<i>Spectrin Inc Cardiovascular Diagnostic</i>
Biomedical/ Life Sciences		Rev1

Company Snapshot: Spectrin's diagnostic kit will be an ELISA-based immunoassay that detects novel cardiac-specific fragments that are uniquely produced during cardiotoxic events, such as treatment with chemotherapeutic drug.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	The technical team is dedicated and capable of achieving the goals. The management team has added support from committed advisors with market expertise to solidify their market strategy and assist with future fundraising.
G	Opportunity/Market Size	Clear unmet need and potential market size in \$ Billions.
Y	Intellectual Property Protection	In negotiations with OSU for rights to a relevant patent application pending in the US and Europe.
Y	Proof of Concept	The connection between the blood marker and chemotherapy-induced heart failure has been demonstrated preliminarily with reagents available today.
Y	Potential Investor/ Business Partner Engagement	Spectrin has identified grant opportunities and venture investment sources. They will initially prioritize non-dilutive over dilutive funding.
Y	Business Model	Creation of an ELISA will enable Spectrin to establish clinical utility of the assay as an adjunctive test to monitor chemotherapy safety.
Y	Growth Plan in Ohio	Team resides in, and is committed to company growth in Ohio.
Y	Project Plan/ Budget Narrative	The plan to develop the ELISA with a well-known assay development
Y	ESP Interaction	The team has re-engaged Rev1 to jointly develop strategy.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Spectrin will develop an ELISA to establish the clinical utility of a marker of chemotherapy-induced heart failure, a common but silent toxicity of cancer therapies. The team will generate data from testing the new ELISA in chemotherapy patients, with the goal of selling the assay as a service to cancer treatment centers. The team will benefit from the newly added pharmaceutical business expert and renewed strategy development activities with Rev1. This application is recommended for funding.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 24-8325	Afference	Amount Requested: \$200,000
<i>Licensing Institution</i>	Case Western Reserve University	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 35	<i>Afference: Enabling touch for the spatial computing age</i>
Software/ Information Technology		JumpStart

Company Snapshot: Afference is developing an electrotactile fingerless glove to be used in spatial computing market, initially targeting hand tracking based virtual reality gaming.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	The two-member team has significant startup and fundraising experience coupled with technical expertise in neural interfaces and artificial perception.
Y	Opportunity/Market Size	Haptic experience is difficult to estimate in gaming industry, 2022 estimates for all gaming devices in AR and 0.25 of VR provide ~\$23B in all hardware and software, with an estimate for haptics of ~\$1B.
G	Intellectual Property Protection	Strong IP position based on granted patents and applications at Case Western Research University.
Y	Proof of Concept	Currently at a TRL5 prototype. TVSF award anticipated to take the prototype to a TRL 6 within 1 year.
G	Potential Investor/ Business Partner Engagement	Received PreSeed funding; connecting with potential new investors for seed funding round. Talking with OEM hardware licensing partners
Y	Business Model	Plan to develop, sell development kits to enterprise customers, then shift to a licensing model. Won CES 2024 Best of Innovation XR tech
G	Project Plan/ Budget Narrative	Plan is consistent with building MVPs to further test sensation quality.
Y	Growth Plan in Ohio	Afference plans to grow the business in NE Ohio.
Y	ESP Interaction	Interacting with JumpStart.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Afference has access to licensing significant intellectual property from Case Western and has one issued patent on IP from work at Afference. The team has significant startup and fundraising experience coupled with technical expertise. The team has received PreSeed funding and plan to transition from providing the haptic devices to licensing OEMs to manufacture and sell the devices. The plan is consistent with obtaining MVPs to further test sensation quality in a gaming market. The platform technology is also applicable to be used as a training tool for various careers, jobs, etc.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 24-8326	Visual Detection Systems, LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	University of Akron Research Foundation	Amount Recommended: \$200,000
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: No	<i>Opioid Detecting Wet Wipe</i>
Biomedical/ Life Sciences	University of Akron Research Foundation	

Company Snapshot: Visual Detection Systems, LLC (VDS) is developing the First Defense Wipe, a wet wipe to identify opioids through a safe, reliable, instantaneous, and “simple” color change.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	Team is committed, capable and experienced. VDS has assembled, and is actively working with a consortium of companies to facilitate successful product development and launch.
Y	Opportunity/Market Size	Initial \$75 million target market for the opioid detection wipe will be the U.S. EMS, police, and corrections officers. Follow-on products will target additional market opportunities.
Y	Intellectual Property Protection	VDS holds an exclusive option to license the patent rights from U Akron and is in active negotiations for an exclusive license. Deep know-how and other IP in development by VDS consortium partners.
Y	Proof of Concept	The color change technology has been demonstrated in a wet wipe format. Currently improving shelf stability, reliability, and production using manufacturing partner's automated production lines.
G	Potential Investor/ Business Partner Engagement	Consortium partners are already investing cash and in-kind to complete the development work and market the final product.
G	Business Model	The consortium partners have the skills and facilities to complete development and launch. Other versions of product are planned.
Y	Growth Plan in Ohio	The partners include four Ohio entities and one from Wisconsin.
Y	Project Plan/ Budget Narrative	Plan is well developed, achievable. TVSF funds to support launch.
Y	ESP Interaction	VDS is engaged with Bounce Innovation Hub and JumpStart.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Visual Detection Systems has demonstrated a ready-to-use, single use disposable wipe that immediately shows a color change indicating the presence of an opioid. First responders have a need for fast, easy opioid detection in the field. Current products are laborious and slow. VDS will finalize chemistry and other conditions to enable manufacturing at scale, and prepare for market launch, expected later in 2024. TVSF funding will support launch strategy, and creation of warehouse and inventory systems. This application is recommended for funding.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 24-8327	Argos Advanced Solutions Inc	Amount Requested: \$200,000
<i>Licensing Institution</i>	NASA Ames Research Center	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	Heart ID
Software/ information Technology		Entrepreneurs' Center

Company Snapshot: Argos is targeting the biometric authentication market to enhance security measures and increase user presence verification.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	A strong Team composed of experienced CEO, CTO, CFO, and COO in startups and fundraising with a solid, active advisory team. Good track record in Business Development and start up fund raising.
G	Opportunity/Market Size	Over \$100B/yr. globally and \$ 34B in US with 22% CAGR.
Y	Intellectual Property Protection	Nonexclusive with NASA, Ames, Iowa Center.
G	Proof of Concept	Since founding company in 2021, Argos, with internal funding of ~\$ 1M, has advanced the technology to TRL 6 with accuracy > 98 % in controlled in house testing.
G	Potential Investor/ Business Partner Engagement	Early discussions with potential investors to secure \$2M in Seed and \$5-8M in Round A investments in 2024 and 2025 respectively.
Y	Business Model	Well-articulated with initial and future target customers/markets. Projected revenue over \$ 30M in 2028.
Y	Project Plan/ Budget Narrative	A well laid out and achievable plan and budget narrative.
Y	Growth Plan in Ohio	Projected staff is 75 in 2028.
Y	ESP Interaction	Active interactions with Entrepreneurs' Center and Dayton incubator.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: A very strong Team with an active Advisory board to introduce the novel breakthrough real time biometric based authentication to potential customers. The market potential is very large and Argos is working with some impactful target customers such as Chase Bank. The Team is recommended to stay focused to gain acceptance of their technology at Chase and in the meantime develop a robust implementable business plan for capturing the full potential of the offerings.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 24-8328	Lyceum AI, INC.	Amount Requested: \$200,000
<i>Licensing Institution</i>	Naval Postgraduate School	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	AI Teaching Assistant
Software / Information Technology		Entrepreneurs' Center

Company Snapshot: Lyceum AI is leveraging a Curated Heuristic Using a Network of Knowledge ("CHUNK") learning system for enhanced and personalized education using a web-based platform.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	The team has experience and capabilities in education, AI and entrepreneurship that are well suited to this opportunity.
G	Opportunity/Market Size	Bringing AI based instruction into the classroom in a way that truly includes the instructor is a significant market opportunity.
Y	Intellectual Property Protection	The team is in the process of licensing a relevant patent application. New filings are also projected.
Y	Proof of Concept	The Proof of Concept MVP is appropriate and ambitious.
G	Potential Investor/ Business Partner Engagement	The founders are well networked in the educational and investor communities.
G	Business Model	Initial customer penetration and acquisition is well considered. A well proven SaaS model is envisioned.
Y	Project Plan/ Budget Narrative	Plan is well conceived and optimistic.
G	Growth Plan in Ohio	Team expressed a strong intent to base and expand in Ohio.
Y	ESP Interaction	Team has engaged with the Entrepreneurs' Center.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: This is a strong team who is well positioned to bring a differentiated AI education product into a large market. The TVSF project is well conceived and has an optimistic scope. The team's background creates confidence that the TVSF project will be successful.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 24-8329	Engineered Products of Ohio, LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	NASA-Glenn Research Center	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>Stealth RFID for Anti-Counterfeit</i>
Sensors	JumpStart/ Team NEO	

Company Snapshot: The company is incorporating NASA's patented nanoionics-based radio frequency (RF) switches to develop non-RF based passive tags for lifecycle tracking and provide resistance to cyber security threats.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	The Team consists of 3 staff with a strong and extensive background and experience in RFID, technical, manufacturing and business development skills. Good contacts with DoD clients and other industrial clients
G	Opportunity/Market Size	Initial target market of Defense and NASA markets are TAM is \$1.65 B; SAM is \$1.24 B; and SOM \$0.93 B
Y	Intellectual Property Protection	Received approval from NASA to license 4 patents in February 2024
Y	Proof of Concept	TRL 6 is based on work done over the last 3-4 years with STTR fundings. Reached TRL 6 with STTR Lewis TOPS funding.
G	Potential Investor/ Business Partner Engagement	In discussions to raise funds by June 2024 in the amount of \$4 M
Y	Business Model	Seems reasonable; perhaps very aggressive. Projected revenue by 2028 is \$ 296.78 M!
Y	Project Plan/ Budget Narrative	Project plan and budget narrative are reasonable and achievable.
Y	Growth Plan in Ohio	Projected staff growth to over 50 by 2027.
Y	ESP Interaction	Working with JUMP Start and Youngstown Business Incubator.
	Evaluator Recommendation	This application is recommended for funding.

	Weak	Meets	Exceeds
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Comments and Recommendations: A small but effective Team with proven track record in RFID field with an ongoing business and access to contract manufacturing. Leveraging very effectively resources from Local Growth Advisors for next round of investments. Has on-going discussions with potential end users for the nano wire-based RFID –Lockheed Martin ventures (LM) and Parker Hannifin. The Team is strongly urged to add needed scale-up, marketing and business development staff to fully realize the potential of the nano wired RFID sensors.

Proposal Summaries - Phase 2 Not Recommended for Funding

Proposal 24-8315	Protein Capture Science	Amount Requested: \$200,000
<i>Licensing Institution</i>	Ohio State University	Amount Recommended: \$ 0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 19,20,21	<i>Preparation of the Next Generation of Protein Purification Platform iCapTag for GMP Production</i>
Biomedical/ Life Sciences		Rev1

Company Snapshot: Protein Capture Science LLC has developed, validated and are selling an intein cap self-cleavable tag, iCapTag™, technology for the protein purification market. The company is focused on introducing this platform into Good Manufacturing Practices (GMP)-driven protein and protein fragment purifications for mid-size and large-scale protein manufacturers.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	Management team, while individually creditable, appears to have significant misalignment on key company issues, including valuation. In addition, proforma year 5 revenues decreasing from \$5.3M (application) to \$2.0M (question answers) does not instill confidence.
G	Opportunity/Market Size	The purification of biologics is a large, high value, growing market.
G	Intellectual Property Protection	The company is expanding upon its already licensed solid IP portfolio.
G	Proof of Concept	The proof of concept is well considered and addresses the key next steps needed to enter the GMP protein purification market.
Y	Potential Investor/ Business Partner Engagement	The company has formed relationships with supply chain partners and has communicated with early-stage investors.
R	Business Model	The supplementary proforma provided is not attractive with \$2.0 M revenue in year 5. Slow revenue growth to date is also a concern.
G	Project Plan/ Budget Narrative	The project plan and associated budget are creditable.
Y	Growth Plan in Ohio	Plans to grow in Ohio were confirmed in supplementary materials.
Y	ESP Interaction	The company has a long-term relationship with Rev1.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Protein Capture Science's product and the associated market are very attractive. The company could benefit from the addition of an experienced commercial leader who identifies and implements options for faster growth and better team alignment.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 24-8319	CrossLIMS, LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	University of Cincinnati	Amount Recommended: \$ 0
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: 35	<i>CrossLIMS: A Collaborative Laboratory Information Management System</i>
Software/Information Technology		CincyTech

Company Snapshot: CrossLIMS integrates data from multiple medical research databases from across an organization into a single, secure, cloud-based e-system for data and workflow management.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	Management team consists of two individuals with data management and health research software development. Team would benefit from the addition of an experienced entrepreneur and commercial leader.
Y	Opportunity/Market Size	Market size is identified as \$2.3 Billion in 2023.
R	Intellectual Property Protection	No patents or trademarks have been filed. No proprietary algorithms identified or claimed. Copyright options being explored with local law firm.
G	Proof of Concept	Demonstration of the MVP during interview indicates a strong working model with significant potential to aid researchers in their work and save time in generating reports, searching through data.
R	Potential Investor/ Business Partner Engagement	No business partner engagement nor investor equity shown in pro forma, however, discussed interactions with angel & other investors.
R	Business Model	Business model, expenses outlined and pro forma do not agree. No assumptions shared regarding misalignment.
Y	Project Plan/ Budget Narrative	Budget and Narrative clarified in pre-Interview questions & interview.
Y	Growth Plan in Ohio	CrossLIMS is committed to maintaining headquarters in Cincinnati.
Y	ESP Interaction	Meeting with CincyTech.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: CrossLIMS has the software development skills and technical knowledge to develop the product to the current stage. The demonstration of the MVP indicates a strong working model with the potential to aid researchers. The team would benefit from the addition of an experienced or serial entrepreneur with a background in software development for healthcare applications.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 24-8320	Shark	Amount Requested: \$200,000
<i>Licensing Institution</i>	Air Force Research Laboratory	Amount Recommended: \$ 0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>Autonomous Rescue Craft</i>
Sensors	Entrepreneurs' Center	

Company Snapshot: Shark Rescue Systems is commercializing an optionally piloted autonomous jet ski with add-on automated airdrop and self-righting capability. Initial target market is military & defense. Other possible markets include law enforcement, search/rescue, firefighting, and lifeguarding.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	The 3- member team consists of CTO/inventor-full time, Rapid Innovation Engineer, AFRA, Center for Rapid Innovation; Business Development-full time, 20-year AF veteran, SERE Specialist; and CFO-as needed, WPAFB, general business/fund raising.
R	Opportunity/Market Size	Potential markets include military and commercial search and rescue applications. These markets appear to be attractive. However, details in proposal are lacking/confusing.
Y	Intellectual Property Protection	Exclusive field-of-use license for three patent applications has been granted from USAF. Patents appear to have broad coverage, but are not public so details are uncertain.
Y	Proof of Concept	Prototypes have been evaluated in field conditions. TRL between 6 and 7. TVSF funds will be used to ruggedize and provide more cost-effective production units for scaled-up manufacturing.
Y	Potential Investor/ Business Partner Engagement	USAF/N market pull with interest in rapid GSA procurement when the jet skis are available in commercial configuration.
R	Business Model	Incomplete/confusing manufacturing strategy (contract/ in-house) and product features. Equity ask appears unrealistic/perhaps excessive.
Y	Project Plan/ Budget Narrative	Vendors identified. Significant in-house manufacturing capabilities.
G	Growth Plan in Ohio	Business plan is to remain/grow in Ohio.
Y	ESP Interaction	EC Dayton working with applicant.
	Evaluator Recommendation	This application is not recommended for funding.

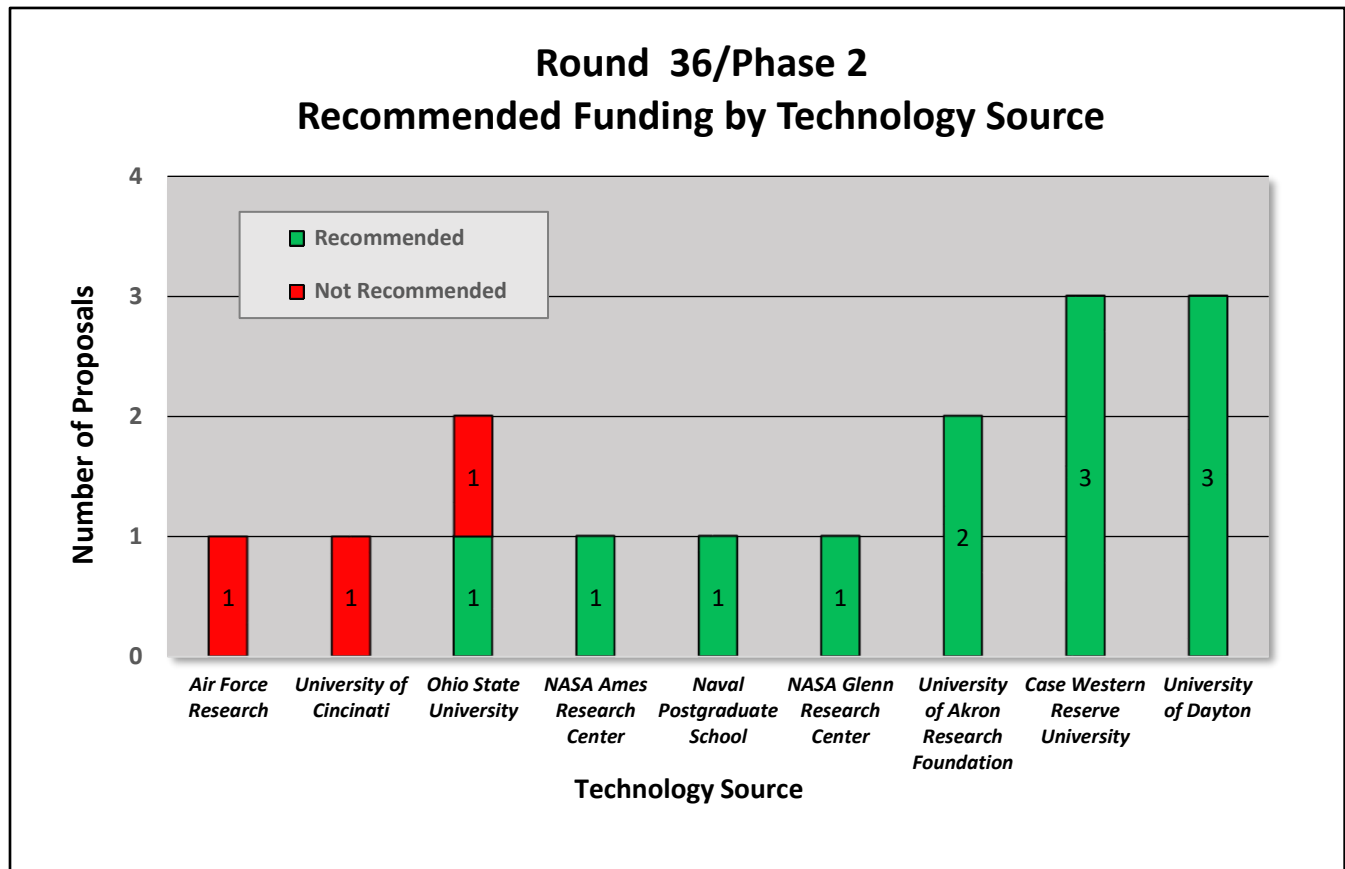
Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Balanced management team having US military background with engineering, operational, and fund-raising expertise. Inventor and SERE operational specialist bring exceptional military contacts & operational awareness to the project. Market applications have high market penetration potential. Proposal states strong level of interest in the military for these products. Incomplete/confusing strategy for in-house versus contract manufacturing of sub-assemblies; i.e., off-the-shelf jet skis, to improve efficiencies/reduce overhead. Equity ask appears unrealistic given investor ROI expectations, and perhaps more than needed if cash flow can be more efficiently managed during the initial growth period.

4) Round 36 Analysis

Figure 1 shows the proposal activity and funding recommendations by technology source for Phase 2 proposals. Both Case Western Reserve University and University of Dayton were the most active technology source with three applicant submissions followed by two applicant submissions from University of Akron Research Foundation and Ohio State University. One submission was received from Air Force Research Laboratory, University of Cincinnati, NASA Ames Research Center, Naval Postgraduate School and NASA Glenn Research Center. All three applications from Case Western Reserve University and University of Dayton, two applicant submissions from University of Akron Research Foundation, and one each from Ohio State University, NASA Ames Research Center, Naval Postgraduate School and NASA Glenn Research Center are recommended for funding.

Figure 1. Round 36 Funding by Technology Source



TECHNOLOGY VALIDATION AND STARTUP FUND

Figure 2 depicts Phase 2 proposal activity and funding recommendations by Third Frontier focus area. In this Round, seven of the fifteen proposals were in Biomedical/Life Sciences (47%), Four of fifteen were in Software/Information Technology (27%), two of the fifteen proposals were in each of Advanced Manufacturing (13%) and Sensors (13%). Six of fifteen in Biomedical/Life Sciences (40%) and three of the fifteen in Software/Information Technology (20%), two of the fifteen in Advanced Manufacturing (13%), and one of the fifteen in Sensors (7%), were recommended for funding. This round represents 53% in Third Frontier Technology areas that were non-Biomedical/Life Sciences. The previous sixteen rounds have averaged 55% of the applications in Biomedical/Life Sciences.

Figure 2. Round 36 Phase 2 Proposal Activity by Third Frontier Technology Area

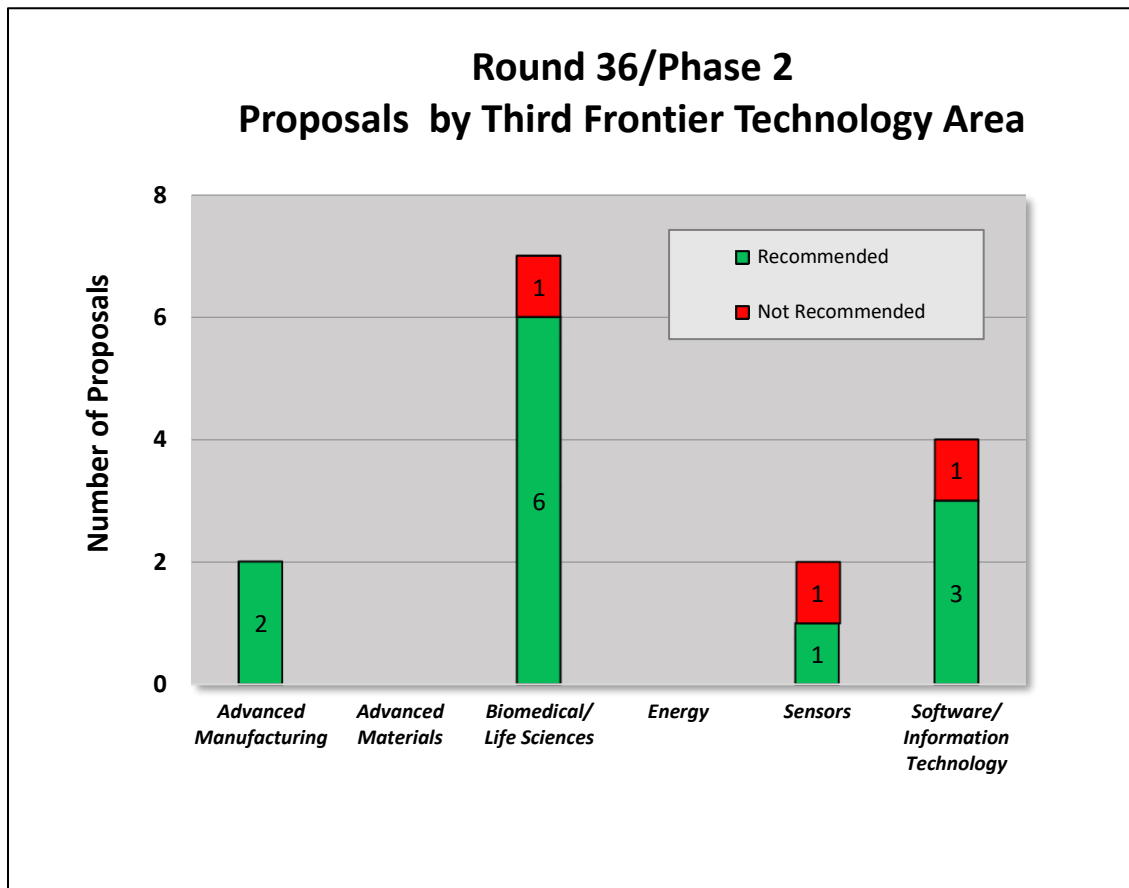
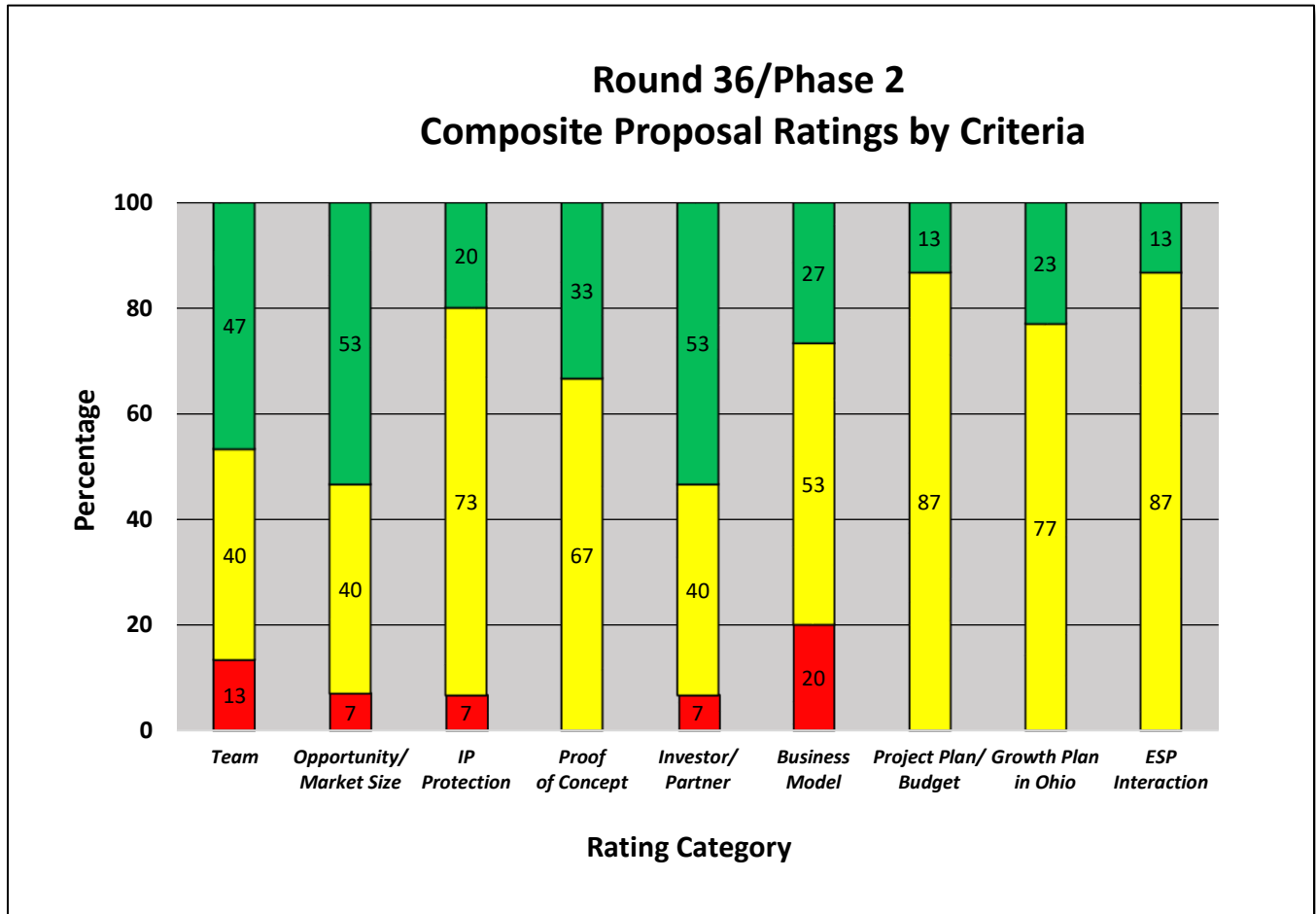


Figure 3 shows the aggregate ratings by evaluation criteria for all Phase 2 proposals. Setting aside the Growth Plan in Ohio and ESP Interaction categories, the Proof of Concept and Project Plan/ Budget Narrative had the lowest percentage of weak scores in this round. Business Model followed by Team had the highest percentage of weak scores in this round.

Figure 3. Round 36 Phase 2 Proposal Rating Summary



In the previous sixteen Rounds, business model was the lowest rating in Rounds 20-23 (53% average $\geq$ meets) and Round 26 (28%). The RFP was revised prior to Round 24 to elicit stronger business models and it appears that the proposals have provided stronger business models in Rounds 24-36 (64% average $\geq$ meets). Round 26 appears to be an anomaly. Future rounds will be monitored to see if the improvement continues.

Figure 4: Rounds 20 to 36 Phase 2 Analysis of Business Model

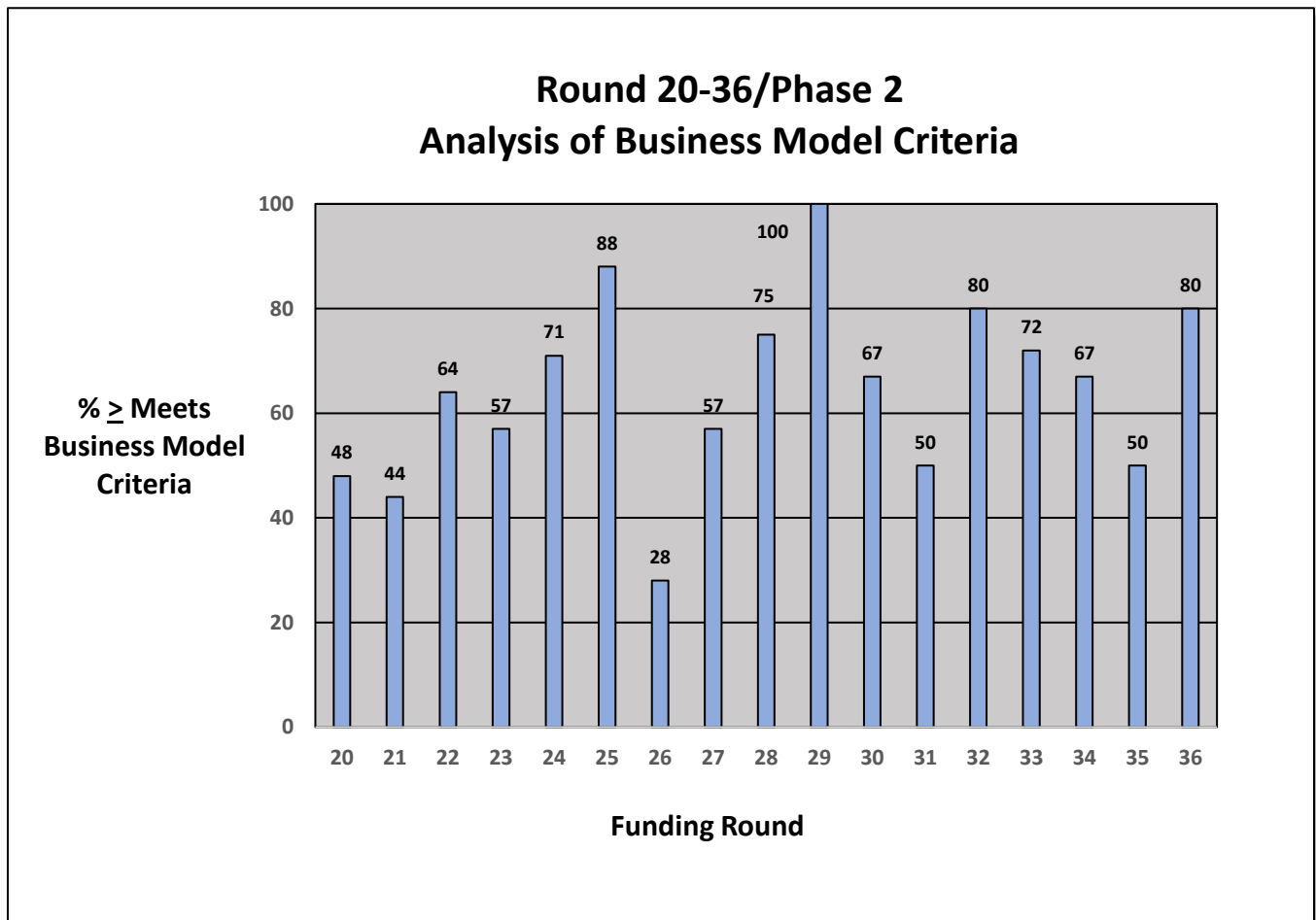
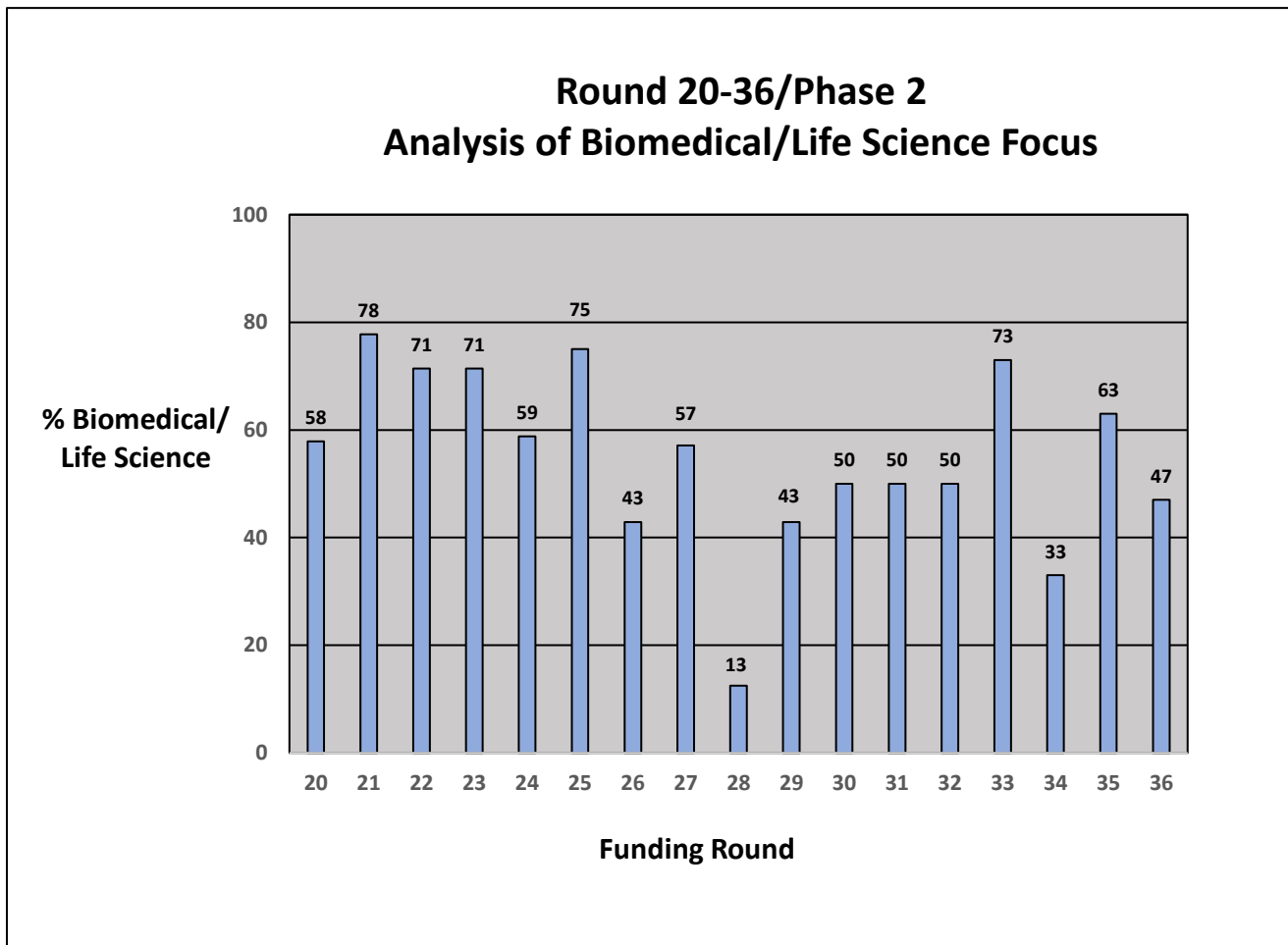


Figure 5 shows the percentage of Biomedical/Life Sciences applications for the last seventeen Rounds. Round 36 represents 47% in Third Frontier Technology areas that are Biomedical/Life Sciences. The previous sixteen rounds average 55% of the applications in Biomedical/Life Sciences.

Figure 5: Rounds 20-36 Phase 2 Analysis of Biomedical/Life Science Focus



Carry Through and Reapplication

Phase 1 Carry Through: There were 5 Phase 2 applicant that previously received Phase 1 funding. Three applications are recommended for funding.

Phase 2: There were 5 reapplications in this Round. Two were first time reapplications and one is recommended for funding. Two were second time reapplications and are both recommended for funding. One was a third time reapplication and is not recommended for funding.

Appendix I

Summary of Redwood team and qualifications

Redwood, as a company, has been providing technology commercialization services for over 9 years while each team member has been active in this field for over 25 years.

Each Redwood team member

- possesses an advanced technical degree and extensive business proficiency
- has worked across the spectrum of technology commercialization from invention to successful market introduction
- understands how to assess a concept case from the perspective of aligning technologies to product applications in specific markets
- has lived, both conceptually and literally, the iterative process of understanding market needs and wants, value chains and who the customers are within the value chain

Team members have all worked for major corporations, research institutions, venture capital firms and technology start-up companies gaining a comprehensive understanding of what is necessary for development teams to successfully commercialize a technology. The Redwood team has served as evaluators for the Ohio Advanced Manufacturing program and an individual team member served as an evaluator for CALF, TIP and IOF loan programs for over a decade.

The five members of the Redwood team are highly qualified evaluators for the TVSF program and have combined experience and expertise in the following areas (combined years):

Commercializing technology into market pulled products (125+ years)

Market/Technology Assessment (140+ years)

Startup/ Spin out companies (50+ years)

Board member/Advisor to Startups (30+ years)

Evaluating/ monitoring RFPs/ Funding selection (40+ years)

The following is a brief summary of the five principal team members used in this evaluation Round.

Herb Bresler

- BS Biological Sciences, University of Maryland; BS Secondary Science Education, University of Maryland; PhD Immunology and Infectious Diseases, The Johns Hopkins University School of Hygiene and Public Health
- Former Senior Research Leader and Chief Scientist for Health and Life Sciences, Battelle Memorial Institute, responsible for evaluation of new technology-based business opportunities, intellectual property development, licensing and tech transfer; created and implemented new metrics to increase returns on discretionary R&D; cultivated approximately 1150 invention disclosures, 900 patent applications, and 120 granted patents, leading to \$52 million company funding
- Recipient of four R&D 100 awards for breakthrough medical devices in neuroscience and diagnostics
- Former Director of the Laboratory of Cellular Immunotherapeutics at the Arthur G. James Cancer Hospital and Research Institute at The Ohio State University

John McArdle

- BE, Manhattan College, MS, Northeastern University, Chemical Engineering
- MBA, Finance / International Business, University of Chicago (Booth School of Business)
- Former Business Development Manager, Battelle
- Former Product Line Manager – Koch Industries
- Former Technical Sales Manager, Allied Signal Corporation
- Recognized expert in water and wastewater treatment technologies
- Successful track record of introducing innovative technologies for a variety of municipal, industrial, and military applications in domestic and overseas markets.

Jim Sonnett

- BS, University of Virginia, MS, University of Massachusetts, PhD, University of Delaware, all in chemical engineering
- Former Vice President – Science and Technology, Battelle Health & Life Sciences
- Former R&D Leader – W. L. Gore & Associates and E. I. DuPont
- Built and led high impact innovation organizations in aerospace, electronics, and life sciences
- Former Board Member – Velocys, Ventaira, Battelle Ventures
- Adjunct professor – Ohio State University Fisher School of Business

Susan Stanton

- BS, Millersville University, Chemistry, MPh, Syracuse University, Organic Chemistry, PhD, University of Rochester, Organic Chemistry
- Personally developed 12+ products and led new product development teams at Mobay, Alcoa & Nexicor
- Holder of 10+ patents
- Former VP Market and Technology Assessment at the National Technology Transfer Center

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- Over 15 years as an angel investor in technology-based startups
- Over 15 years as an evaluator for Ohio Third Frontier funds including IOF, CALF and TIP
- Over 8 years teaching market and business analytics to STEM graduate and post doc students.

Bhima Vijayendran

- BS, University of Madras, MS, University of Madras, PhD, University of Southern California in Polymer and Surface Science, MBA, University of New Haven
- Former Senior Research Leader and Vice President Business Development, Battelle Memorial Institute; Chief Research Officer, Battelle Science and Technology, Malaysia
- Former Director, Discovery Research, PPG Industries
- Recognized as one of the leading authorities on advanced materials, special chemical and polymer systems in numerous markets including: Renewable and clean technology, Energy, Nano Technology and Industrial Products.
- Recipient of ten R&D 100 awards and over 100 patents and numerous other awards.

Appendix 2

TVSF objectives and phases

The Technology Validation and Start-up Fund (TVSF) provides grants under two phases to transition technology from Ohio Eligible Research Institutions into the marketplace through Ohio start-up companies. Under Phase 1, Ohio Research Institutions may apply for a pool of funds to support validation/ proof that will directly impact and enhance both the commercial viability of their unlicensed technologies and ability to support a start-up company. Under Phase 2, Ohio start-up and young companies may apply for funding to commercialize a technology they intend to license from a university or an Ohio research institution.

The goals of Phase 1 include:

- Generate the proof needed to move technologies to the point that they are either ready to be licensed by an Ohio start-up company or deemed unfeasible for commercialization. The institutions are encouraged to work with potential Ohio licensees to identify the proof needed.
- Perform validation activities such as demonstration and assessment of critical failure points in subsequent development, prototyping, scale-up and commercialization in order to generate this proof with strong preference for these activities being performed by an independent 3rd party source.

The goals of Phase 2 include:

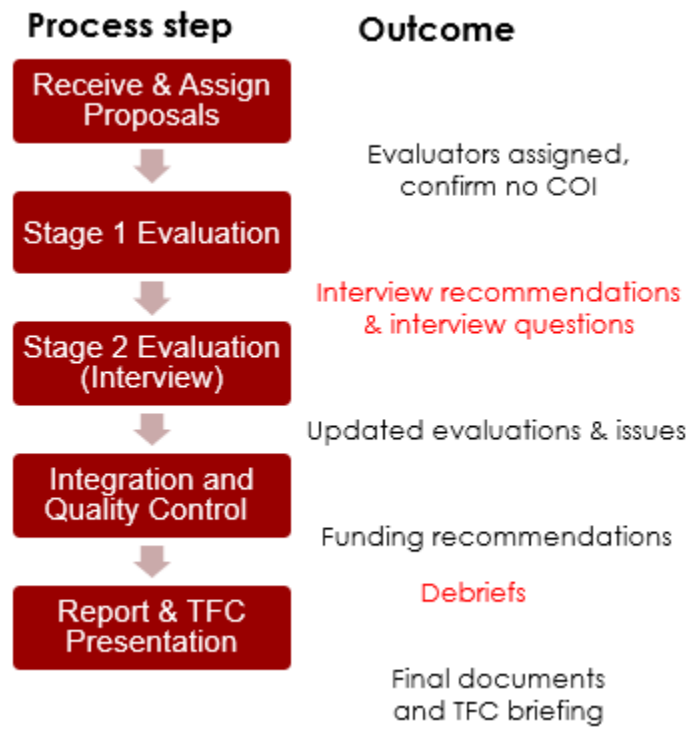
- Accelerate the commercialization of technology by Ohio start-up companies that license technology developed at Eligible Institutions during the critical early stage of life of the company.
- Generate the proof needed to move technology to the point where it is able to be commercialized or additional funds for commercialization can be raised. A clearly identified path to subsequent funding opportunities and working directly with potential investors to define the proof needed for investment into the company is strongly encouraged.
- Funded activities may include, but may not be limited to, beta prototype development and deployment to potential customers for testing and evaluation and market research/ business development in order to generate the proof needed.

Based upon these goals, the proposal evaluation criteria were developed. The proposals were then evaluated based on the criteria.

Description of review process

Review summary. Our overall review process flow and outcomes by stage are shown in Figure 1. A similar process has been successfully used by Redwood in prior projects for public and private clients. Discussions were held with the TVSF program manager after all but the initial step in Figure 1.

Figure 1. TVSF Evaluation Process



Review and Assign Proposal In this first step proposals were summarized and a primary evaluator was assigned who has the appropriate background and no conflict of interest.

Stage 1 Evaluation Stage 1 evaluations were conducted for each proposal using the criteria shown below in Tables 1 and 2. Differentially weighted criteria were used to evaluate Phase 1 and Phase 2 proposals. Each proposal was rated on a 0 (absent) – 5 (Outstanding) scale for each criterion, an approach used by the NSF and in other State of Ohio programs. The weightings reflect the experience of the Redwood team and our belief that some factors, for example team and market opportunity in Phase 2, are more important than others.

The entire review team subsequently discussed all the evaluations to ensure consistency and agreed upon which applicants to invite for interviews. Interview questions were then provided in advance to each applicant.

TECHNOLOGY VALIDATION AND STARTUP FUND

Stage 2 Evaluations (Interviews) The standard procedure for this step is: In-person, 45-minute interviews were held with each invited applicant to discuss the advance questions plus other topics of interest to the evaluators. Two Redwood team members participated in the interviews in person with additional team members joining via Zoom or a conference call.

Integration and Quality Control Proposal evaluations were updated based on interview results. A calibration review was held by the review team to ensure that evaluations were performed consistently and that any changes made were a result of team consensus. Based on this review, proposals were recommended for funding.

Table 1 – Phase 1 Evaluation Criteria

Criterion	Weighting	Description
Alignment and Compliance	Go / No go	Institutional alignment with TVSF intent and compliance with RFP
Project Selection Committee	20	Skills, background and commitment of the committee members
Deal Flow; Budget Strategy	15	Is the projected deal flow consistent with the requested budget to enable committing funds within 1 year?
External Participation	15	Does process ensure validation activities will be performed by 3 rd parties; ESPs and state-funded programs/organizations are enlisted to enhance commercialization activities of the project?
Track Record	15	Is there a strong Phase 1 or comparable program track record of licensing and newco creation? If not, is there a plan for improvement?
Metrics	15	Realism and impact of proposed metrics, including licensing, start-ups.
Project Management & Experience	15	Is there a strong project management strategy and appropriate experience of people who allocate the pool of funds and manage individual projects?
Project Selection Process	5	Is there a clear, appropriate process for project selection?

TECHNOLOGY VALIDATION AND STARTUP FUND

Table 2 – Phase 2 Evaluation Criteria

Criterion	Weighting	Description
Alignment & compliance	Go / No Go	Proposal alignment with TVSF intent and compliance with RFP
Management Team	20	Skills, background and commitment
Opportunity / market size	15	What is the market segment and total addressable market? Is it a platform or breakthrough technology or incremental improvement? If breakthrough, is it compatible with viable commercialization pathways?
IP Protection / License	15	Is IP adequately protected, does it enable the business model, is it differentiated from likely competition, is license likely within 9 months?
Compelling Proof of Concept	15	Was meaningful input from potential customers and key performance metrics used to design Proof of Concept? Are the competitive advantages compelling for potential customers?
Potential Investor / Business Partner Engagement	10	Is there company engagement / collaboration independent of licensing institution, including financial backing?
Business Model	10	Is the business model realistic AND achievable? Can the service / manufacturing model be scaled?
Project Plan / Budget Narrative	5	Is the budget consistent with proof in 1 year?
Start-up in Ohio	5	Does a start-up exist or is it planned? Will the start-up be in Ohio?
ESP Interaction	5	Is team engaged with ESP? Has team incorporated feedback from ESP into the project, proposal or business plan?